

The empirical recurrence for OEIS A297717: recovering a conjecture that is not written down

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Abstract

OEIS A297717 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 160$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297717 is “Number of $n \times 5$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1, 2 or 4 neighboring 1s.”. It has offset 1 and begins

12, 258, 2717, 36981, 512354, 6812856, 91994155, 1242370107, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Jan 04 2018 21:57:00, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 160$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 160$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 320$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	3	c_{23}	47215431808357	c_{45}	−58867526402283279	c_{67}	−37965781427152543
c_2	65	c_{24}	−85854003046602	c_{46}	22232306732530181	c_{68}	8620109597525530
c_3	974	c_{25}	−372659146265787	c_{47}	169331763226716417	c_{69}	18494587424474240
c_4	2559	c_{26}	−384482998249747	c_{48}	−88841568286773539	c_{70}	−2681840214349148
c_5	−6672	c_{27}	634317645484519	c_{49}	−346204799739577450	c_{71}	−6813233383191569
c_6	−190573	c_{28}	2374135336343227	c_{50}	−23361314423321120	c_{72}	1465424633744359
c_7	−713401	c_{29}	1914627407840762	c_{51}	559254616067039643	c_{73}	1748851869416999
c_8	−438712	c_{30}	−3464592630241349	c_{52}	259529287002934040	c_{74}	−1282695052737217
c_9	16946521	c_{31}	−9956819099538304	c_{53}	−632335872132504554	c_{75}	299490978610408
c_{10}	73217887	c_{32}	−4864126068148344	c_{54}	−635193391701701435	c_{76}	50135219906278
c_{11}	94076795	c_{33}	13999119118180517	c_{55}	431712007459249093	c_{77}	−86744612041332
c_{12}	−854286377	c_{34}	25804245527780770	c_{56}	774142455311130725	c_{78}	25782290481372
c_{13}	−3992049167	c_{35}	−295835456577939	c_{57}	−12281366028555536	c_{79}	−19341790881768
c_{14}	−5759123116	c_{36}	−39139124048432458	c_{58}	−702397800197862900	c_{80}	6623994274688
c_{15}	27136547751	c_{37}	−31959207319715375	c_{59}	−99517230628573416	c_{81}	1060857654768
c_{16}	131515133086	c_{38}	38461827257361144	c_{60}	400045302979487525	c_{82}	508534794352
c_{17}	187454197889	c_{39}	63003789051426297	c_{61}	268823913115448814	c_{83}	230010604736
c_{18}	−574591003561	c_{40}	−17238386308431333	c_{62}	−260539610278019977	c_{84}	−56402411936
c_{19}	−2788344950454	c_{41}	−93534673168029088	c_{63}	−170249239536786085	c_{85}	6407365696
c_{20}	−3719614070168	c_{42}	−24729413502170232	c_{64}	38630371713837123	c_{86}	−2486058368
c_{21}	8372770779265	c_{43}	92163028079186634	c_{65}	114140336968661111	c_{87}	18191360
c_{22}	39288268243022	c_{44}	29489510540696167	c_{66}	−28807457897821363		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

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